



Asthma Guide

Qena STEM school

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Abstract

Working to eradicate public health issues (diseases) and expanding Egypt's industrial bases are two of the country's present major difficulties, and they both merit our consideration for this project. Raising industrial bases and eliminating health problems are extremely complex challenges that impact many other aspects of life, such as Egypt's economy and health. Since impairments and chronic illnesses are related to this health concern, we must find a smart system solution to enable those who suffer from chronic illnesses live more comfortably. In order to contribute to the problem's resolution, the chronic illness "asthma" was selected. The idea was to build a hardware system with two sensors (MQ135 and DHT11) to measure the percentage of smoke, temperature, and humidity in the air—the main factors that directly affect asthma patients. This hardware system would then be connected to a mobile application that would contain an icon for information and instructions, as well as additional icons that would allow the patient to select which environmental conditions—hot >35°C, cold <15°C, or both—trigger him in regard to temperature, as well as whether to turn on an alert (flash) if those conditions are met. Additionally, the system would display readings in an OLED and an application through a Bluetooth module to inform the patient. then the system response time to the input and processing which was 0.5sec and accuracy of sensing the environment were tested "design requirement achieved" After the project was completed, it was tested by keeping an eye on the temperature throughout the day and noting the decisions made by the mobile application when it reached <15 and >35.Also, the alerts had been turned on for humidity and the percentage of smoke that was identified. Ultimately, by controlling environmental influences, asthma patients will become more self-reliant and knowledgeable about their illness and its management.



Introduction

Egypt faces many issues that have a detrimental effect on the country's social life, economy, ecology, and health; these issues stand in the way of Egypt's advancement, growth, and prosperity. Among the difficulties facing our nation are expanding Egypt's industrial base and eradicating public health problems and diseases. But the elimination of public health difficulties and diseases is one of the most pressing challenges that has a definite detrimental effect and given rise to numerous major issues. The increase in percent of people affected with chronic diseases is because of negligence in eradicate public health issues (see fig 1)Many projects and prior solutions have been applied to overcome this problem such as: The Butterfly iQ which is a handheld ultrasound device that utilizes AI technology to aid in the diagnosis and monitoring of various medical conditions, Aidoc is a radiology platform that utilizes AI technology to assist in the detection and prioritization of critical findings in medical imaging studies. It aims to enhance the efficiency and accuracy of radiologists and Google DeepMind's Streams is an AI-powered platform designed to assist clinicians in monitoring and managing patients with acute kidney injury (AKI). In addition to the challenges that Egypt faces, there are also specific health issues that require urgent attention. Asthma is one of these problems; it is a chronic respiratory condition that affects millions of individuals globally. Due to its high rates of death and prevalence, asthma is a serious public health concern in Egypt. Recent data indicates that 15% of Egyptians have asthma, and that the country has one of the highest asthma-related death rates worldwide. Contributing to the improvement of asthma sufferers' lives by developing a hardware system that works with a mobile application software system. Since various persons are impacted by various circumstances, our system's application depends on three variables: temperature, humidity, and smoke. Thus, the design specifications that we shall address measuring the environment's temperature and humidity, measuring CO2 levels, and testing the system's reaction time in relation to its input and processing. Patients with asthma will be able to better manage environmental influences, become more self-reliant, and learn more about their illness and management techniques which is the solution objective.

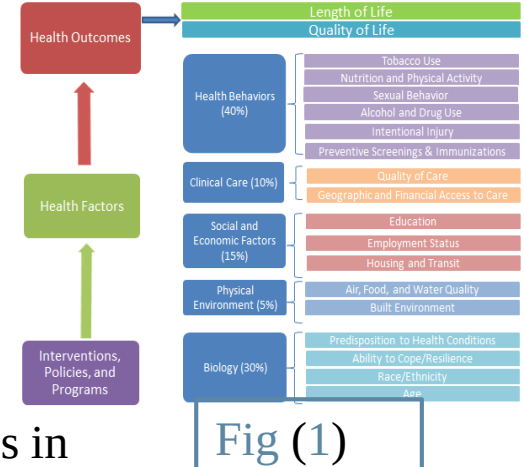


Fig (1)



Materials & Methods

Item	Quantity	Usage	Source of purchase	Picture
DHT11	1	To detect the temperature and humidity of environment	Electronic shop	
MQ135	1	To detect co2 gas	Electronic shop	
OLED SSD 1306	1	To detect the reading of co2 and temp	Electronic shop	
Wires	30	For communications	Electronic shop	
Nano	1	Give order to sensors	Electronic shop	
ESP8266	1	Communicate with applications	Electronic shop	
Bluetooth module	1	Communicate Nano with application	Electronic shop	

Methods:

Following the inclusion of the materials required to construct the prototype, this part will detail the building process utilizing the previously specified materials.

The steps used in construction of the prototype:

-First, as shown in fig.(2), the connections between the sensors (MQ135 & DHT11) and the ESP3266, Arduino Nano board, and OLED were completed.

Connecting the MQ135 sensor to an Arduino Nano to measure the percentage of smoke and the DHT11 sensor to an ESP to measure the temperature resulted in the data from the sensors being communicated to mobile applications over Bluetooth and Wi-Fi, respectively.

-Next, a template of the forearm was created so that the hardware system could be built onto it and the patient could use it easily (see fig 3). Subsequently, the hardware system was linked to a mobile application by Bluetooth and Wi-Fi. The application features icons that enable patients to regulate variables that the hardware "sensors" identify, as illustrated in fig (4).

Test plan:

The test plan called for using Bluetooth and Wi-Fi to transfer the data from the two sensors—the MQ and DHT11—to a mobile application. In order to identify whether the warning is turned on or not. following that ,Positive, negative, and observational data were gathered.

- Compiling the code and obtaining data by using an OLED, mobile application, and serial monitor to identify issues.

- After testing the system's response time to input and processing, which was 0.5 seconds, it was discovered that utilizing an ESP8266 in addition to an Arduino Nano would support the system and speed up processing. This indicates that the "design requirement" was met.

- The system was verified by continuously monitoring the temperature, triggering an alert "flash" when the temperature reaches below15°C or beyond 35°C and when the humidity reaches above 65%.

- Additionally, the smoke from candles, which contains gases (CO& CO2) and particulate matter, was used to test the accuracy of the environment sensing (design requirement achieved) by checking for an increase in smoke percentage to more than 50 ppm.



Fig (2) connection between sensor



Fig (3) forearm

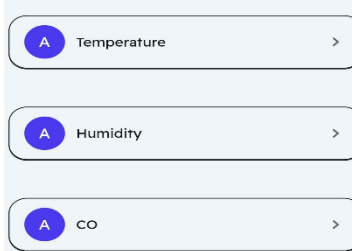


Fig (4) mobile application



Results

The system performed well with the sensors' mutual interference, indicating that the materials chosen for its construction were sound ones.

The smoke produced by lighting the candle is displayed on the OLED in parts per million (ppm) (see table 1).

Table (1) CO2 readings

Test condition	CO2 Level in ppm
Before lighting the candle	43
After lighting the candle	281

The graph (1) shows the Level of CO2 in the air detected by MQ135

With respect to time intervals, the temperature changes over the course of a whole day (24 hours) has been recorded by providing the readings on the webpage via ESP8266.

_Two triers have been made with only difference in the option of the icon of temperature trigger condition on the mobile application. With (day one 1 "Hot" icon, day 2" cold" icon).

_The change of humidity during a whole day intervals have been recorded with the readings of the temperature in day 2

Table(2): shows the that had only flash turned on one time in the whole day 1

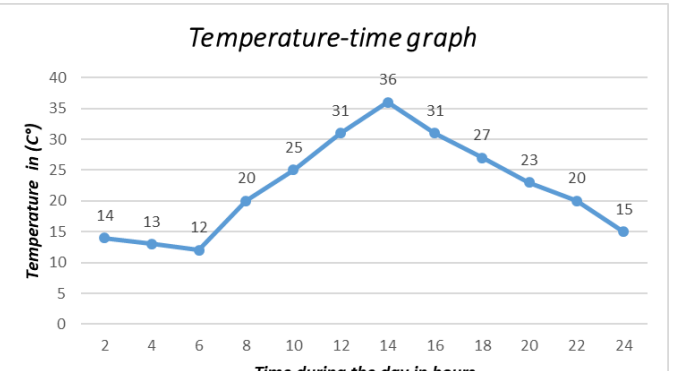
Time during the day " Hour"	Temperature" C"
2	14
4	13
6	12
8	20
10	25
12	31
14	36 "Flash turned on"
16	31
18	27
20	23
22	20
24	15

Table(3): shows that the flash had turned on two times in day 2

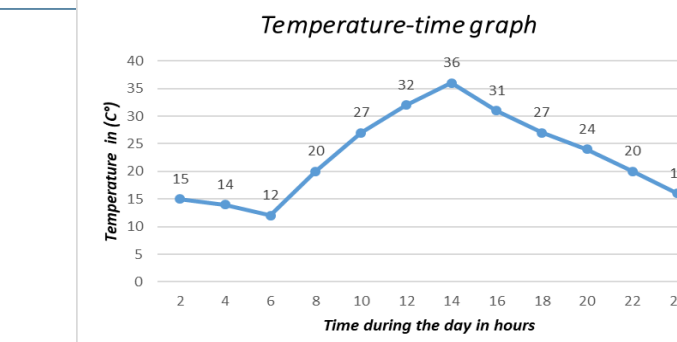
Time during the day " Hour"	Temperature" C"
2	15
4	14
6	12
8	20
10	27
12	32
14	37
16	31
18	27
20	24
22	20
24	16

Table 4: shows that the flash had turned on two times during the early morning

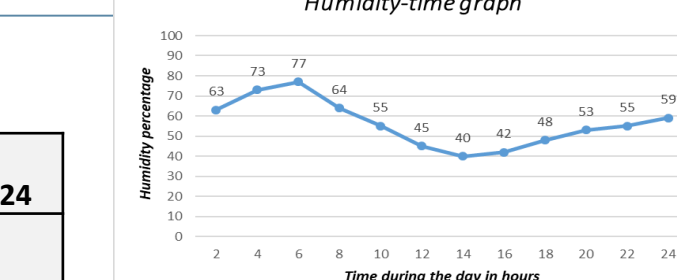
Time during the day " Hour"	Humidity percentage (%)
2	63
4	73 "Flash turned on"
6	64
8	55
10	45
12	40
14	42
16	48
18	53
20	55
22	59
24	



graph (2) shows the change in temperature every two hours during day 1



graph (3) shows the changes in temperature every two hours during day 2



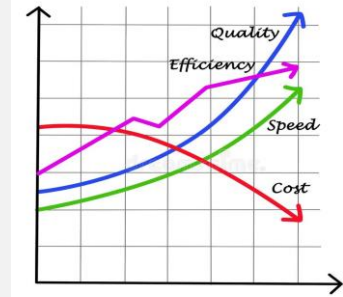
graph (4) shows how humidity changes u during the day



Analysis

Understanding an issue is the first step towards solving it. The primary goal is to broaden Egypt's industrial base by developing strategies and instruments that enhance public health care and make the lives of individuals with chronic diseases easier. The first step in fixing this kind of issues is identifying its root cause. Given that issues of this nature are global in nature. Resolving these issues would be comparable to finding a hidden gem. Which means that the main goal of all of this is to construct a system that satisfies a set of predetermined constraints.

The Arduino Nano's slower processing speed and inefficiency when handling multiple sensors connected simultaneously caused the project's original results to fall short of expectations. As a result, the plan was changed to utilize the ESP8266 board in conjunction with a DHT11 sensor that was connected via Wi-Fi to a mobile application. To detect smoke particulate, the Nano has merely been attached to the MQ135 device. And the system displays good results in a shorter amount of time which was 0.5 second after the two boards are connected and the system interferes with a mobile application. which, when the processing system's performance grows, make the project more efficient (see the opposite graph 5).Initially, the Arduino Nano was linked to the MQ135 to detect smoke. This was checked by burning a candle and placing a glass cup over it to make sure smoke didn't escape into the air. To assure safety, the MQ135 was placed with the candle underneath the cup (see fig 5)



graph (5) represents the elements that affect the efficiency of the project

. The desired findings from the sensor are displayed in graph (1). The graph indicates that the sensor measured 43 ppm of CO2 before the candle was even lit. This is because earlier attempts to read the sensor failed due to systematic mistakes in the code, which were fixed with the assistance of Talk GPT (an AI model). And after lighting the candle, there was a noticeable rise in the CO2 level, reaching 281 ppm. The reading on the OLED indicated this change, as shown by the fig(6,7). This data was also sent via the Bluetooth module to the mobile application to warn when the readings surpass 50 ppm, which is what occurred the second the candle was lit; the flash activated as a result of the abrupt change in air quality brought on by the rise in CO2 levels to 281 ppm

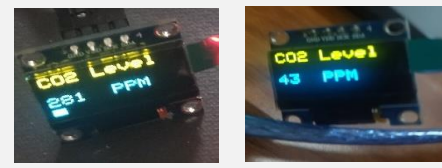


Fig (6,7) CO2 readings (smoke)

. The second factor was to measure the temperature and humidity because these two variables have a significant impact on people with asthma. Some people are affected by increases in temperature, while others are affected by decreases in temperature, and some are even affected by any deviation from the normal range. Additionally, the concentration of water vapor in the air must be taken into consideration, making the temperature and humidity variables both important. Regarding this, the DHT11 sensor was also connected to the Arduino Nano. However, it was later discovered that it is difficult and inefficient to connect the two sensors to the Nano; as a result, the ESP8266 was used to connect to the DHT11, and because of this it was easier to use Wi-Fi via the ESP8266 and retrieve the data via the IP (see fig.8),Which was connected to the application through Wi-Fi to enable it to take action according to the readings .The temperature readings throughout the course of the two test days are displayed on the graph (2&3). (Table 2) demonstrated that the flash had been triggered at 14:00, which was expected given that this is the time of day when the sun is at its hottest and the mobile application is set to trigger a flash when the temperature rises above 35 C, taking into account that the hot triggering condition has only been enabled in the mobile application. In contrast to the events of day one, (Table 3) shows day two the flash was activated to draw attention only when the temperature dropped below 15 degrees Celsius between 4 and 6 in the morning. And the reason for this is that the "cold" icon trigger condition had been triggered, which was predicted because the lowest temperatures often occur right before sunrise. (The application's figure 6 illustrates the condition).Because of the high concentration of water vapor in the morning, the result of the humidity during the day demonstrates its inverse relationship to the temperature. This highlights the importance of monitoring humidity levels throughout the day because it directly affects asthma patients and can be predicted based on the relationship between temperature and humidity. The peak of humidity occurs at 6 am, when the temperature is at its lowest. (see graph 6). (Figure 10) clearly illustrates the building of the communication system, and the flow map on the other side provides a summary of the preceding process's analysis and procedure.

_Many laws and principles was used to build the system and accomplish all wanted design requirements. Here are some main reasons why the candle's smoke has been chosen to make test on: 1-Particulate Matter: Burning a candle produces tiny particles, often referred to as particulate matter. This can include both visible soot particles and invisible particles like PM2.5 (particles with a diameter of 2.5 micrometers or smaller). 2-Water Vapor: As the wax combusts, it reacts with oxygen to produce carbon dioxide (CO2) and water vapor (H2O).



fig (10) shows the components of the prototype

A statistical measurement of the strength of a linear relationship between two variables called The correlation coefficient has been used to calculate the relation between the temperature and humidity through this law: (H: Humidity & T: Temperature) $r = \frac{\text{Cov}(H, T)}{\sqrt{\text{SDH} \cdot \text{SDT}}}$ $r = \frac{n \cdot \sum xy - \sum x \cdot \sum y}{\sqrt{(n \cdot \sum x^2 - (\sum x)^2) \cdot (n \cdot \sum y^2 - (\sum y)^2)}}$ $n=3$ $\sum x = 24+20+16=60$ $\sum y = 53+55+59=167$ $\sum xy = (24 \times 53) + (20 \times 55) + (16 \times 59) = 3316$ $\sum x^2 = (24^2) + (20^2) + (16^2) = 1232$

By substitution $r = -.0981$

Therefore, the correlation coefficient for these pairs of data is -0.981, indicating a strong negative correlation between temperature and humidity.

Learning outcomes:

Chemistry L.O5: The Ideal Gas Law ($PV = nRT$) can be related to the behavior of gases, including CO2. It describes the relationship between pressure (P), volume (V), temperature (T), and the number of moles of gas (n). While the project may not directly apply this law, understanding the principles of gas behavior was useful in interpreting CO2 sensor readings.	Physics inventory: Thermodynamics (Heat Transfer): Monitoring temperature involves principles of heat transfer. Understanding how heat is transferred (conduction, convection, radiation) was interpreting temperature changes and implementing triggers based on temperature conditions.
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Physics L.O4: - Wi-Fi Communication (Electromagnetic Waves): The communication between our devices using Wi-Fi involves principles of electromagnetic waves. While this was not directly related to asthma, understanding the basics of wireless communication was essential for implementing IoT devices.	Biology L.O1: - The mechanism by which synapses work in the human body, and when and how reflexes occur, is the primary source of inspiration in the way the project was built. As the system consists of a group of devices connected to each other and connected to a mobile application, which gives decisions in the same way that the nervous system works
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Chemistry L.O1: different types of analysis had been used like: quantitative & qualitative analysis, the project to ensure its effectiveness and meet our design requirements.



Conclusion

Asthma project is a potential solution that deals with a serious health issue in Egypt, according to the study of the solution and the prior data results. The project provides a comprehensive system that allows asthma patients to monitor and control environmental elements that trigger their symptoms by utilizing contemporary technologies. The project gives patients tailored advice and alerts using its hardware gadget, AI algorithms, and mobile application, enabling them to properly control their illness. Apart from its health advantages, it is a safe and extremely sustainable solution. By lessening the influence of ambient factors on asthma attacks, the project's design complies with all safety regulations, guaranteeing that it does not present any risks to users or the environment.



Recommendations

- The following suggestions are for anyone wishing to enhance this project or for future work on it:

1. Considering other asthmatic triggers, such as pollen or air pollution, in order to improve asthma patients' quality of life and make life easier for them. This would increase the efficacy of treating asthma symptoms.
2. The accuracy of identifying environmental triggers for asthma patients is increased when a dedicated smoke sensor is used in place of the MQ135 for smoke.
3. Expanding the functionality of the mobile application Working together with medical professionals: In order to enable medical personnel to give their asthmatic patients better individualized and knowledgeable care, think about creating a framework that would allow them to access and examine the data gathered by the system.
4. Including an icon for input from medical professionals and asthma patients to better understand their individual requirements and preferences for asthma management within the system.

All things considered, it is advised to put our proposal into action since it satisfies numerous design criteria that show its efficacy and quick processing and response time in comparison to most IoT communication systems.



Literature citation

1- *Asthma Symptoms*. (2023, January 26). Retrieved from Asthma & Allergy Foundation of America: <https://aafa.org/asthma/asthma-symptoms/>

2-*DHT11 Sensor Definition, Working and Applications*. (2019, August 6). Retrieved November 16, 2023, from EIProCus: <https://www.elprocus.com/a-brief-on-dht11-sensor/>

3- Halliday, D. (2004). *Physics*. Wiley Custom Services.

4- Heinecke, R. (2020, April 7). *Air Quality Sensor*. Retrieved from Breeze Technologies: <https://www.breeze-technologies.de/air-quality-sensors/>

5- staff, M. c. (2022, March 5). *asthma*. Retrieved October 29, 2023, from Mayo Clinic: <https://www.mayoclinic.org/diseases-conditions/asthma/diagnosis-treatment/drc-20369660>

6- Urry, L. A., Cain, M. L. 1., Wasserman, S. A., Minorsky, P. V., Reece, J. B., & Campbell, N. A. (2017). *Campbell biology*. Eleventh edition. New York, NY, Pearson Education, Inc.



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